

FRIEDRICH BEST'S CASE Z. WITH MISIDENTIFICATION OF OBJECT ORIENTATION

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In 1917, the German ophthalmologist Friedrich Best¹ published in 'von Graefe's Archiv für Ophthalmologie' a report on 38 brain-damaged patients with visual field defects due to gunshot wounds or grenade splinters. Working in a field hospital close to the front line, Best saw the patients within days of their being wounded and observed them for the next 2 to 4 weeks during the course of their disease. The aim of his study was to describe the neuropathology and symptomatology of hemianopia in full detail. Best basically included patients with retrorolandic lesions affecting the primary visual cortex, the optic radiation, or the lateral geniculate body. His description offers a valuable insight into the attempts to understand the nature of symptoms associated with visual field defects at the beginning of this century.

In the first part of his report, Best explains some problems of measuring the visual field in patients with hemianopia, describes the autopsy findings in selected patients and gives an overview of contemporary theories about the cerebral location of the macula. Best's main focus was the study of disturbances of visual localization. He viewed the occipital lobes as essential for localizing an object in space with respect to the body. Therefore, he expected to find disturbances in line bisection and in pointing tasks. He found that only patients with complete unilateral homonymous hemianopia show the typical bisection error previously described by Axenfeld (1894) and Liepmann and Kalmus (1900) with the longer segment of the line in the unaffected hemifield, whereas patients with incomplete or bilateral visual field cuts bisect lines atypically with the longer segment in the blind hemifield. Additionally, Best was interested in studying visual agnosia (Seelenblindheit), agraphia and alexia. Moreover, he was concerned with the question whether the patients are able to identify several objects simultaneously. In addition, Best tested whether or not reflexive eye movements are affected by visual field defects.

At the end of the section on visual localization, Best describes one patient in more detail. Case 38, named Z., suffered a gunshot wound through the head 34 days before he was first seen by Best. Z.'s visual fields could not be precisely determined because he could not maintain fixation. Still, Best estimated that 40° in all directions were spared. Z. had no alteration in size perception (dysmetropsia) and no visual hallucinations. While his writing was not affected, the patient was not able to read more than single letters. Z. had no visual object agnosia; he easily recognized any stimuli presented to him. However, once Z. moved his eyes from the stimulus, it took him a long time to find it again leading to difficulties in describing the stimuli in more detail. Z. had no problems in colour perception, but when two or more colours were presented simultaneously, Z. could not recognize them at the same time. When different colours were presented horizontally, he reported only the rightmost colour. When different colours were vertically arranged, Z. found only the one at the top. Best made the same observation with stimuli other than colours. Z. was always aware of only one of several simultaneously presented items. Best did not use a specific term for these latter observations but one probably has to interpret them as simultanagnosia.

Further, Best conducted some mental imagery tasks. While Z. could recall his house or faces of his relatives, he could not remember certain parts of the topography of his home town. Moreover, he was not able to picture in his mind the place where he has been wounded. Best interpreted this deficit as a loss of mental imagery for spatial locations and recent events whereas mental imagery for colours and earlier events was spared.

In another experiment, Best tested for visual and auditory localization of objects with respect to the body. He presented objects and sounds left, right, above or below Z.'s body. The patient was not able to report verbally the location of visual stimuli with respect to his body, and his errors were unsystematic, indicating that he suffered from a complete loss of visual

egocentric localization. Interestingly, he was able to determine verbally the location of sounds with respect to his body and he correctly reported the locations of pressure phosphenes. In contrast, when pointing to objects he usually pointed too low and his gaze accordingly deviated downwards. Moreover, Z. was not able to determine the direction of movements of objects.

To summarize, Z. had no visual agnosia, no difficulties in size perception, no agraphia, and no disturbances of colour perception, but he was not able to locate objects correctly with respect to his body, he could not determine the direction of movement, and he showed a deficit of mental imagery for spatial locations. Moreover, he suffered from simultanagnosia, optic ataxia, and deficits of exploratory eye movements. Thus, one may cautiously conclude that Z. exhibited a combination of symptoms that has been termed 'Balint Syndrome' occurring with bilateral parieto-occipital lesions (Balint 1909, Rafal 1997).

In addition to these disturbances, Best made a further, peculiar observation in patient Z. He presented figures, pictures of persons (e.g. the German emperor at that time; Fig. 1), and other stimuli either in their canonical upright orientations or rotated through 180°. Z. was able to recognize any stimuli presented to him even when they were oriented upside down. In sharp contrast, he was not able to determine the orientations of these stimuli. When Best presented a hand pointing up, down, left or right, Z. could identify the hand but was not able to judge the direction of pointing. Accordingly, when Best presented letters, Z. could not differentiate between the letter 'n' and the letter 'u'. Training of the tasks did not improve Z.'s deficit.

----- Figure 1 near here -----

At the fourth day of testing, Z. developed a fever. In spite of repeated draining of right-sided abscesses his condition deteriorated. Soon he could no longer move his left arm or

leg. Six days later the patient died. According to the autopsy, the point of entry of the projectile affected the upper portion of the left angular gyrus (Fig. 2). The medial exit of the projectile was in the precuneus. The medial point of entry of the projectile into the right hemisphere was again the precuneus, affecting the parieto-occipital sulcus. The lateral exit of the projectile was slightly above the right angular gyrus in the posterior part of the inferior parietal lobule. Along the path of the bullet through the right hemisphere, an abscess was found which nearly extended to the corpus callosum and the lateral ventricle. In addition, purulent meningitis of the right convexity was discovered.

----- Figure 2 near here -----

Although there remain uncertainties concerning the interpretation of the various observations in Z., Best described a clear dissociation in this patient between a preserved knowledge of object identity and an impaired sense of object orientation. Since this classical investigation, six further patients showing the same dissociation have been reported. Solms et al. (1988) asked a patient (WB) with bifrontal abscesses and episodes of inverted vision to sort letters according to whether they were presented in their upright orientation or inverted through 180°. Although WB could read and identify the letters in both orientations, he incorrectly classified their orientation in 35% of the upright presented and in 82% of the inverted stimuli. A comparable observation was reported from a patient with bilateral parieto-occipital lesions (Robertson et al. 1997). The authors presented two letters either upright or inverted. While patient RM correctly named the letters on every trial except one, he incorrectly classified their orientation in 14% of the upright presented and in 64% of the inverted stimuli. Turnbull et al. (1995, 1997) reported three further cases. One patient (LG) suffered from multiple strokes emanating from an arterio-venous malformation in the right

temporo-parietal region; the other two patients had strokes in the right parietal (NL) and the right temporo-parietal area (SC). When the patients were asked to indicate the canonical upright orientation of those objects that they perfectly identified in any of four possible orientations (0° , -90° , $+90^\circ$, 180°), the three patients performed 57%, 53% and 13% incorrect orientation judgements. Karnath et al. (2000) studied a patient (KB) suffering from an old left frontal lesion and new strokes involving bilaterally the parieto-occipital region. They presented pictures of letters, objects, animals, and faces with unambiguous canonical upright orientations in four different orientations (0° , -90° , $+90^\circ$, 180°). KB showed no impairment in identifying the stimuli irrespective of their given orientation. In sharp contrast, she was not able to judge the orientation when the stimuli were presented in a non-upright orientation.

Of some of these patients it was reported that they rotated objects by 90° or 180° in copying (Solms et al. 1988, Turnbull et al. 1995, 1997) or in writing with wooden letters (Karnath et al. 2000). Rotational errors in copying tasks were observed in even more patients suffering from cortical lesions but, unfortunately, their knowledge of object orientation and object recognition was not formally compared (Pillon 1981, Turnbull 1996, Solms et al. 1998).

Best (1917) interpreted Z.'s discrepant ability between determining object orientation and identifying these objects as a kind of 'agnosia' and as evidence that the brain can achieve object recognition without viewer-centered information, i.e. without knowledge of the orientation of the object with respect to the subject. He wrote 'This case demonstrates that the neural centers for processing the spatial features of visual objects are sharply separated from those centers that process the recognition of all other features of these objects' (p. 122). With reference to Z.'s autopsy, he stated that the center for processing object orientation is localized

in the precuneus and adjacent parts of the superior parietal lobule on the hemisphere's medial surface.

Turnbull et al. (1995, 1997) interpreted the phenomenon quite similarly to Best (1917). The authors likewise assumed an orientation-dependent and an orientation-independent route to object recognition in humans. They argued that this view can be accommodated within the two visual systems account of Milner and Goodale (1995). The latter assumed a ventral and a dorsal stream of processing information about the properties of objects and their spatial locations. While the dorsal stream provides the instantaneous and egocentric features of objects, the ventral stream permits the formation of perceptual and cognitive representations which embody the enduring characteristics of objects. Turnbull and co-workers (1995) interpreted the observation of normal object recognition with an impaired sense of object orientation as an operation mediated via the view-independent ventral stream in the absence of the dorsal stream that carries orientation information. Accordingly, the patients showing this dissociation have been considered as agnostic for object orientation.

The interpretation given by Best and Turnbull and colleagues corresponds with viewpoint-invariant theories of object recognition. These hold that the mental representations of previously seen objects are stored in object-centered frames of references based on the objects' geometry (Marr and Nishihara 1978, Corballis 1988). According to these theories, object recognition does not require mental transformations to align the respective image of an object with a represented view of it. The latter has been assumed by viewpoint-dependent theories of object recognition. These theories hold that the mental representations of previously seen objects are stored in a viewer-centered frame of reference determined by the location of the viewer in relation to the object. Accordingly, objects might be recognized by

interpolating between previously seen and stored views (Bülthoff and Edelman 1992) or by transforming the input and/or the stored view (Shepard and Cooper 1982, Tarr 1995).

Support for the notion of a distributed view-based representation of objects in which neurons become tuned to the features present in certain views of an object comes from the recent study of patient KB (Karnath et al. 2000). The results argue that patients with normal object recognition but with misidentification of object orientation are *not* generally agnostic for object orientation. When KB's abilities to perceive objects' canonical upright vs. non-upright orientations were directly contrasted using a forced choice paradigm, the authors found that her knowledge of upright orientation was perfectly preserved but she was not able to judge orientation when the stimuli were presented in a non-upright orientation, i.e. when they were tilted by -90° , $+90^\circ$, or 180° . These data showed that beyond the obvious discrepancy between the ability to identify objects and to determine their orientation, there was also a discrepancy between the knowledge of upright orientation vs. non-upright orientation of objects. Although not systematically tested, observations in the previously studied patients have been reported indicating that these patients, like KB, also were able to gain access to at least some information about objects' canonical upright position.

Thus, it seems that the patients' ability to determine the objects' upright orientation is more robust against neuronal damage than the ability to determine non-upright orientations. This finding of preserved knowledge of upright orientation of objects contradicts Best's (1917) and Turnbull et al.'s (1995, 1997) view of an absence of a neural system that selectively processes orientation information of objects, and argues against their interpretation of a general 'agnosia for object orientation' in these cases. The observation that the patients demonstrate a weakness or inability to determine an object's orientation specifically when that object is presented in a non-upright orientation rather supports the thesis of Perrett et al.

(1998) that one does not need mental rotation to recognize familiar objects in non-canonical orientations. Perrett and colleagues proposed that objects are represented by neurons tuned to view, orientation, and size and that the number of tuned neurons for a particular orientation depends on the amount of experience with it. Since most natural objects are oriented in a gravity-based upright orientation we can assume that many more neurons are recruited to code an object in that orientation. Karnath and co-workers (2000) argued that this numerical bias is sufficient to explain why the patients with impaired sense of object orientation were not generally agnostic for object orientation but rather showed a dissociation between their knowledge of objects' canonical upright orientation and their ability to perceive objects' orientation in non-upright positions. A larger number of neurons tuned for upright orientation would predict that the perception of objects' canonical upright is more robust against neuronal damage or may recover earlier than knowledge of other orientations.

Our ability to recognize objects from many viewpoints is remarkable. The question of how object recognition copes with changes in view, scale, and orientation and how such processes are represented in the human brain are relevant to various fields of cognitive neuroscience. To conclude, it seems that Best's observation of a dissociation between normal object recognition but impaired sense of object orientation in a patient with bilateral parietal lesions opened a new field in human neuropsychology that promises to contribute helpful new insights to our understanding of the neurobiological and the cognitive mechanisms of object recognition.

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Footnote

¹Friedrich Best was born in 1871 in Wermelskirchen, Germany. He received his degree in medicine from the University of Heidelberg in 1894 and wrote his habilitation thesis in Giessen about a type of corneal opacity in 1900. Best expressed wide scientific interests, publishing articles on pathological anatomy, neuropathology, neuroophthalmology and sensory physiology. In 1901, he discovered a new stain which differentiated retinal rods and cones, named 'Best's carmine'. In 1905, while working as a senior physician in Giessen, Best published a paper about an autosomal dominant, slowly progressive, macular degeneration which still bears his name ('Best disease'). Only recently, in 1992, the gene locus for Best's macular degeneration was discovered. Best moved to Dresden in 1906 and became the head of the Department of Ophthalmology at the Friedrichsstätter Hospital. After World War II, he moved to Marsberg and started an ophthalmological practice. He finally retired in 1960 at the age of 89 and died on 6 June 1965.

Figure legends

Fig. 1:

Picture of the German emperor, Wilhelm II, presented upside down.

Fig. 2:

Reconstruction of Z.'s brain lesion according to the autopsy findings.

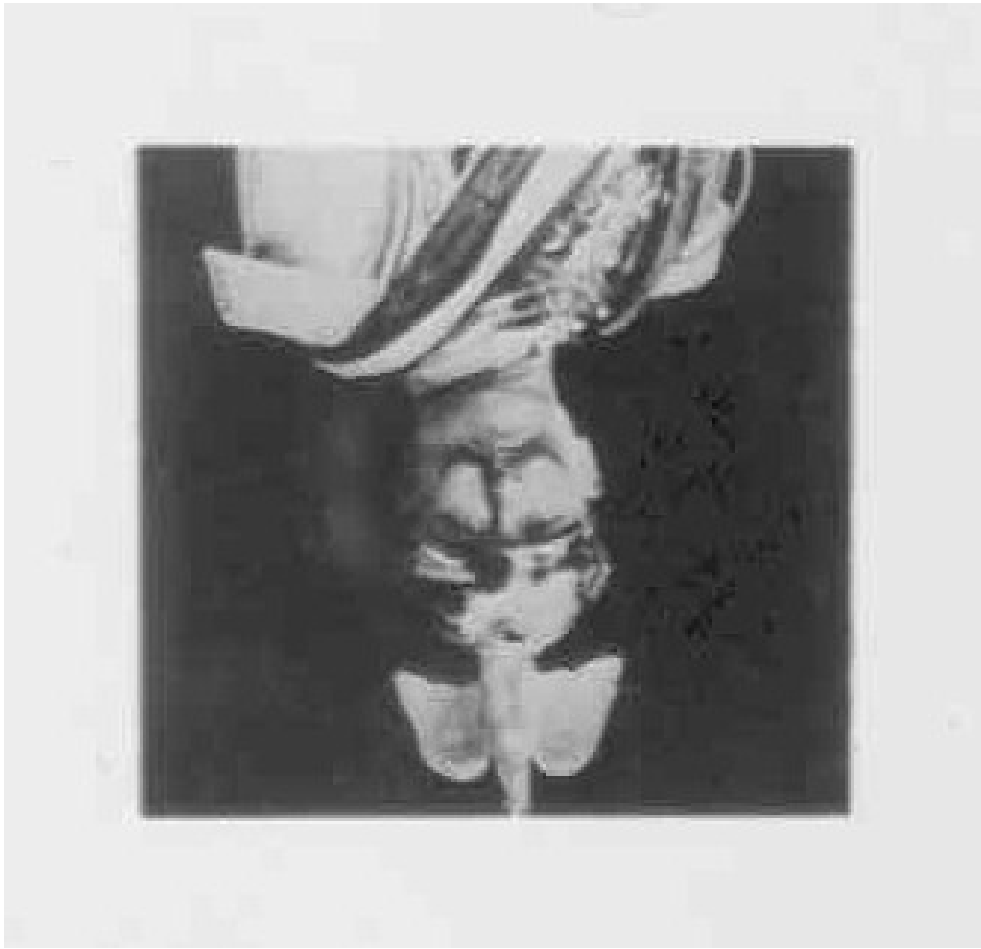


Figure 1

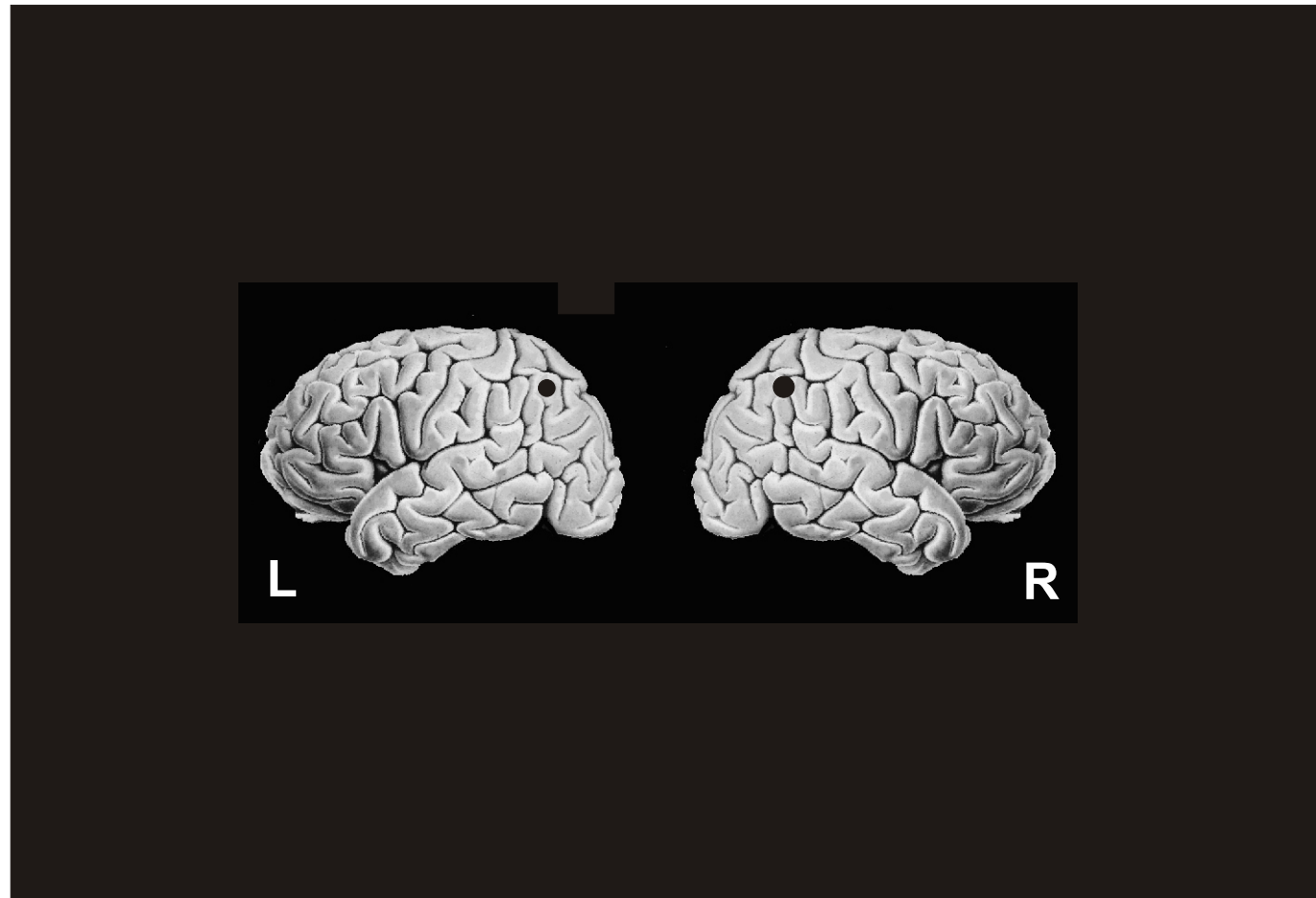


Figure 2